

**A
PROJECT REPORT
ON
ONLINE VEHICLE MANAGEMENT SYSTEM**

Submitted to



SAVITRIBAI PHULE UNIVERSITY OF PUNE

For Partial Fulfillment Of

BACHELOR OF COMPUTER SCIENCE

Submitted By

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&

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Under the guidance of

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Department Of Computer Science

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R.B. Narayanrao Borawake College, (Autonomous) Shrirampur.

In the

Academic Year

2024-25



Rayat Shikshan Sanstha's

R. B. Narayanrao Borawake College, (Autonomous)

Shrirampur – 413709 , Dist - Ahilyanagar



CERTIFICATE

2024-25



“Education through self help is our motto” – Karmaveer

DEPARTMENT OF COMPUTER SCIENCE

Date : / / 2025

This is to certify that **Miss.Ghodke Namita Vinayak** and
Miss.Bankar Mangala Arvind Of T.Y.B.Sc (Computer
Science) has completed project report on
“Online Vehicle Management System”.

Mrs.Shinde K.R.
(Project Guide)

Prof.Dr.Shinde B.S.
(Head of Dept.)

Internal Examiner

External Examiner

*** ACKNOWLEDGEMENT ***

It is distinct honour and proud privilege to acknowledge with our attitude, the keen interest taken by **Mrs. Shinde.K.R.** & inspiring suggestion and constant supervision and encouragement who had made it possible of pursue and complete this project of **"ONLINE VEHICLE MANAGEMENT SYSTEM"** effectively. We would take this opportunity to express our profound series in definers to all the members of staff of their constant help.

It has given us immense pleasure to associate ourselves with through provoking & absorbing. Many thanks to **Prof. Dr. B.S. Shinde Sir (H.O.D of Computer Science)** for his encouragement and valuable guidance.

We greatly appreciate and express our gratitude to all staff members for their keen interest and number of fruitful suggestions at various stages during the completion of this endeavour, which help to reach the depth of this work.

And last but not least, we would like to thanks all our friends for their support and timely help.

Your 're Sincerely,

Miss. Ghodke Namita Vinayak

Miss. Bankar Mangala Arvind

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➤ **Introduction to System**

1. Introduction

The **Vehicle Management System** is an online platform designed to facilitate the selling of vehicles efficiently. This system is developed using **Java** and aims to provide users with a seamless experience in browsing, selecting, and purchasing vehicles.

It serves as a bridge between vehicle sellers and buyers, ensuring a secure and user-friendly transaction process. The System helps sellers list their vehicles with detailed information, and buyers can browse, compare, and purchase vehicles effortlessly.

Traditionally, buying or selling a vehicle requires visiting showrooms, negotiating prices, and dealing with paperwork. This system simplifies all these process and by eliminating the need for physical visits, this system makes the entire process faster, more efficient, and accessible to a wider audience.

2. Scope of System

Scope of Online Vehicle Management System is a web based platform designed to help people buy vehicles easily. It allows sellers to list their vehicles with details such as price, images, and specifications, while buyers can search for and compare different vehicles before making a purchase. The system provides a user-friendly interface where transactions can be done securely without needing to visit physical showrooms.

The platform is built using Java, JSP, PostgreSQL and Apache Tomcat, making it a reliable and scalable system. Administrators manage the platform by verifying vehicle listings, monitoring transactions, and ensuring smooth operations.

Overall, the system simplifies the vehicle buying and selling process, making it more accessible and efficient for both buyers and sellers. It saves time, reduces costs, and provides a convenient way to trade vehicles online.

3. Need of System

- **Convenient Buying & Selling** – Allows users to buy and sell vehicles online without visiting physical showrooms.
- **Wider Reach** – Sellers can showcase their vehicles to a larger audience beyond local markets.
- **Time - Saving** – Eliminates the need for in-person visits, making the process faster and more efficient.
- **Easy Search & Filtering** – Buyers can search for vehicles based on brand, model, price, and features.
- **Secure Transactions** – Ensures safe and reliable payments through integrated payment gateways.
- **User - Friendly** – Provides a simple and easy-to-use interface.
- **Verified Listings** – Ensures only genuine vehicle listings are displayed.
- **Cost-Effective** – Reduces the need for expensive advertisements and showroom visits.
- **Real-Time Updates** – Buyers and sellers receive instant notifications about new listings and inquiries.
- **Quick Communication** – Buyers and Sellers can directly interact through the platform.
- **Better Management** – Admins can monitor transactions and maintain system security.

4. Existing System

- **Traditional Buying & Selling** – People buy and sell vehicles through physical showrooms or local advertisements.
- **Limited Reach** – Sellers can only connect with buyers in their local area.
- **Time-Consuming Process** – Buyers need to visit multiple locations to find the right vehicle.
- **Manual Transactions** – Payments and paperwork are mostly done offline, which can be slow.
- **Lack of Proper Filtering** – Finding a vehicle with specific features requires more effort.
- **Higher Costs** – Seller spend money on advertisements, and buyers may pay extra due to middlemen.
- **No Centralized System** – No single platform where buyers and sellers can interact easily.
- **Limited Reach** – No proper verification of vehicle details or seller authenticity.

5. Proposed System

The aim of Proposed System is to develop a system of improved facilities. The Proposed System can overcome all the limitations of the Existing System. The Existing System has several disadvantages and many more difficulties to work well. The Proposed System tries to eliminate or reduce these difficulties up to some extent. The Proposed System will help the user to reduce the workload and mental conflict.

The Proposed Online Vehicle Management System provide a user-friendly and efficient platform for buying and selling vehicles online. With this system, the vehicle buying and selling process becomes more convenient, cost effective, and secure, eliminating the need for intermediaries and reducing transaction time.

➤ **System Analysis**

1. Feasibility Study

The analysis of current system is most important thing in the development of any new system or project. Feasibility study provides the process for analysis of the current system. The Feasibility Study for the system ensures that the project is practical, cost-effective, and beneficial for users. It examines the system from different perspectives, including **Operational, Economical, Technical, Legal, and Schedule Feasibility** to determine its viability.

❖ **Operational Feasibility :**

The system will enhance user experience by offering an intuitive interface where sellers can easily list vehicles, and buyers can search, filter, and purchase vehicles effortlessly. As in this system, Admin is only Seller, so Admin control ensures platform security by verifying listings and preventing fraudulent activities. It will automate and optimize the vehicle trading process, making it faster and more efficient compared to traditional methods.

❖ **Economical Feasibility :**

The project requires an initial investment for software development, hosting, and maintenance, but these costs are balanced by revenue opportunities such as listing fees, premium ads, and transaction commissions. By reducing the need for physical showrooms, both buyers and sellers save money, making the system a cost-effective solution in the long run.

❖ **Technical Feasibility :**

The System will be developed using Java (JSP, Servlets), PostgreSQL, HTML, CSS, and JavaScript, ensuring a robust and scalable architecture. It will be hosted on an Apache Tomcat Server, making it accessible through web browsers on both desktops and mobile

devices. The use of modern database management, security protocols, and payment gateways ensures efficient data handling and secure transactions.

❖ **Legal Feasibility :**

The system must comply with data protection laws (e.g., GDPR) to ensure user privacy and security. Proper terms and conditions should be established to prevent fraud and define user responsibilities. Secure payment gateways must be integrated to protect financial transactions. Ensuring vehicle authenticity and legal ownership verification is essential to avoid illegal sales. The system should also follow e-commerce and consumer protection laws to maintain trust and legal compliance.

❖ **Schedule Feasibility :**

The System must be developed within a realistic timeframe to ensure timely deployment. The project can follow a structured timeline, including requirement analysis, design, development, testing, and deployment, typically taking 3 to 6 months depending on complexity and team size.

2. Hardware and Software Requirements

❖ Hardware Requirement –

- Processor : Intel Core i3
- RAM : 8.00 GB
- SSD : 512 GB

❖ Software Requirement –

- Operating System – Windows 11
- Frontend – HTML , CSS , Bootstrap & JavaScript
- Backend – Java (JSP & Servlets)
- Database – PostgreSQL Database
- Server – Apache Tomcat
- Browsers – Google Chrome, Mozilla Firefox

3. Fact Finding Techniques

At the time of system or before starting actual work system, analyst has to collect information about the manual system from user which is the system is going to be developed.

In order to develop this information, system analyst prefers any one of the following fact finding techniques or prefer one or more fact finding technique according to the situation.

These techniques are as follows :

❖ Record Review :

Information related which system may be present in form of various record like documents, magazines, letters, etc. This kind of record provides very valuable information to the analyst about the system about organization and various procedure and rules. Record review may be performed at the beginning of the study to collect initial information or at the end of study to compare actual operation.

❖ Observation :

If information cannot be collected through fact-finding technique, then observation method is used. Observation allows getting the information which may be useful for getting some advanced enhancement that other fact finding techniques can't do. With the observation we obtained the first hand information about how activities are carried out and this is useful when we need to actually observe how documentations are handled, how processes are carried out and whether specified steps are actually followed or not.

❖ Interviews :

Interview technique are used for collecting information from individual or from groups. It is fact finding technique where by the system analysis collect information from

individual face-to-face interaction. This was the best technique in which collected maximum information. The necessary question asks to this agent & finding out the application information. The interviewer must plan in advance and should fully know the problem for under consideration.

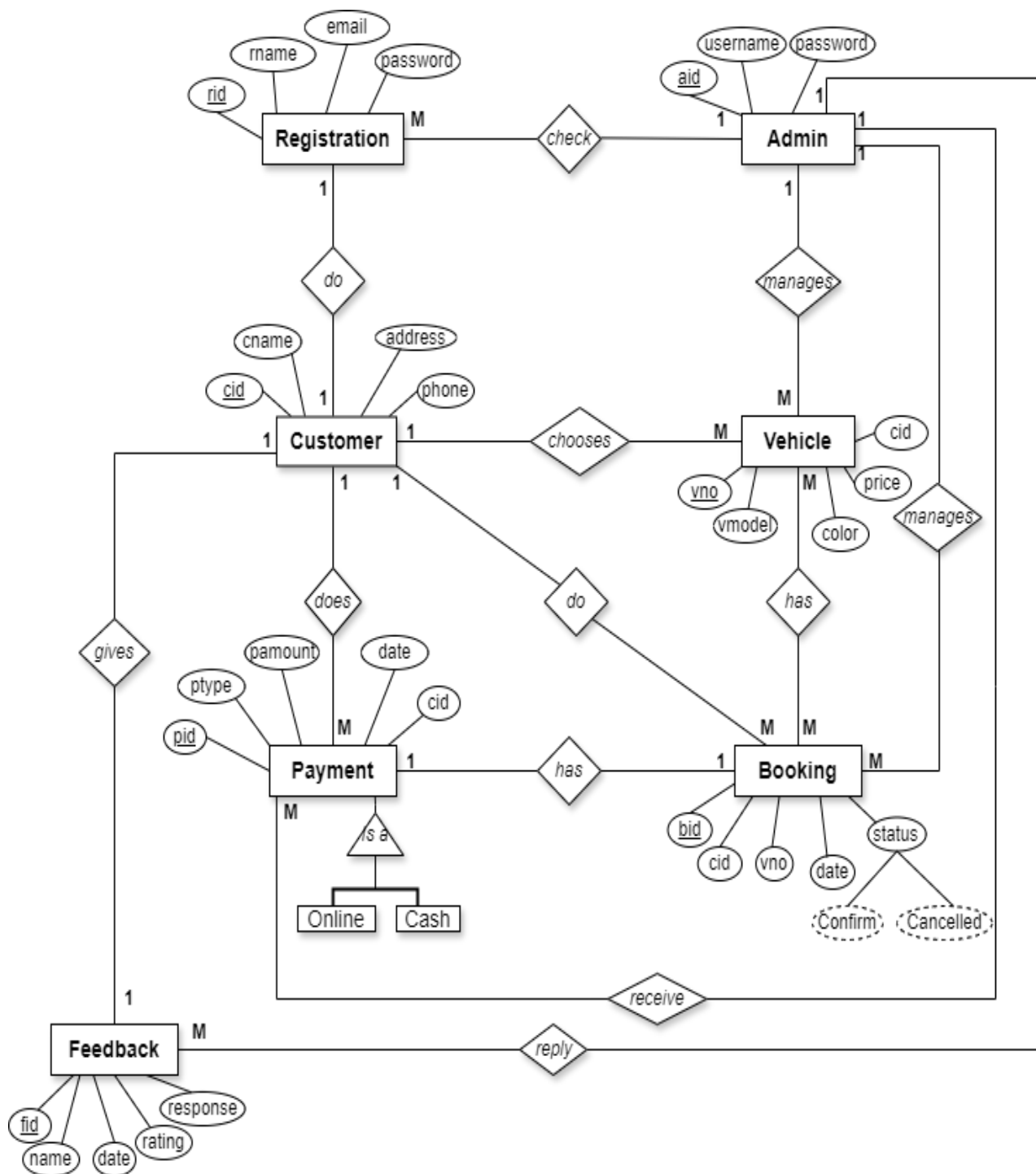
❖ **Questionnaires :**

A questionnaire is the sequence of techniques collect information from user. The questions are clean and simple and to the point we design skillful & imaginative question which we have helped in collect good data regarding our project. In questionnaires system we asks the question likes :

1. What is the primary purpose of your Online Vehicle Management System ?
2. What key features would you like to see in a Online Vehicle Management System ?
3. How often do you require maintenance reminders ? (e.g., monthly, weekly, annually)
4. What is your preferred platform for accessing the system ?
5. What security measures are important to you in a Online VMS ?

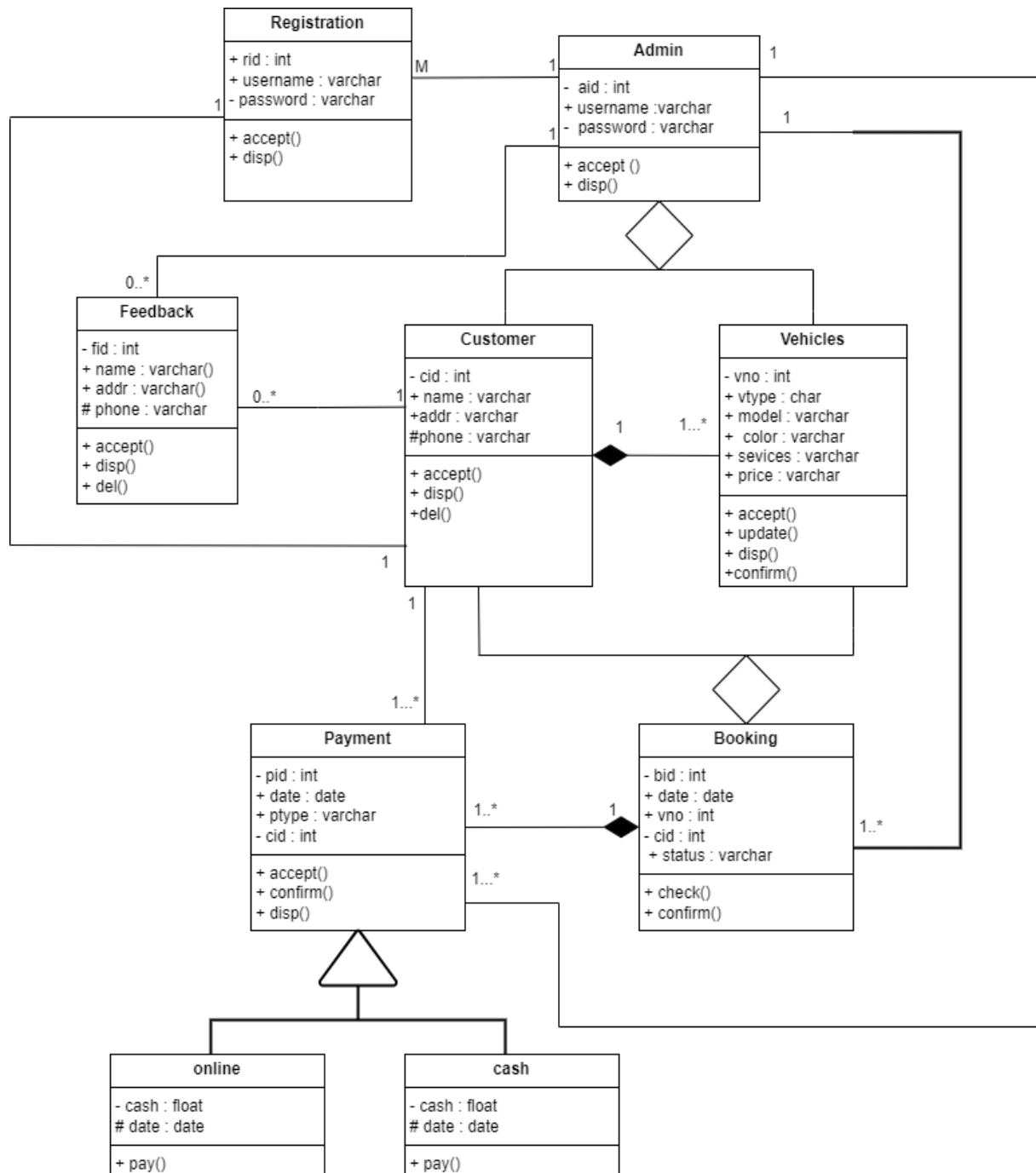
➤ System Design

1. E-R Diagram

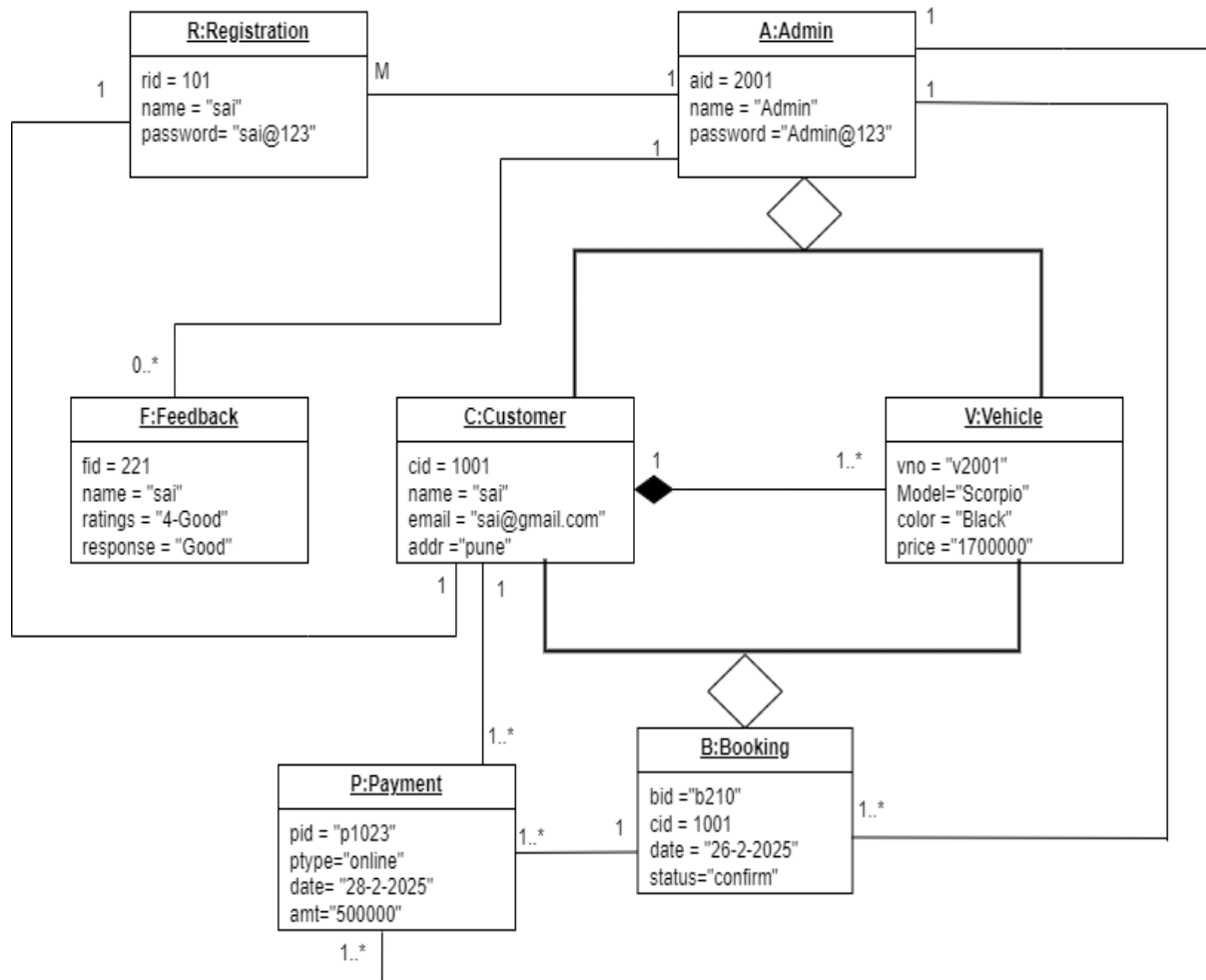


2. UML Diagrams

A) Class Diagram

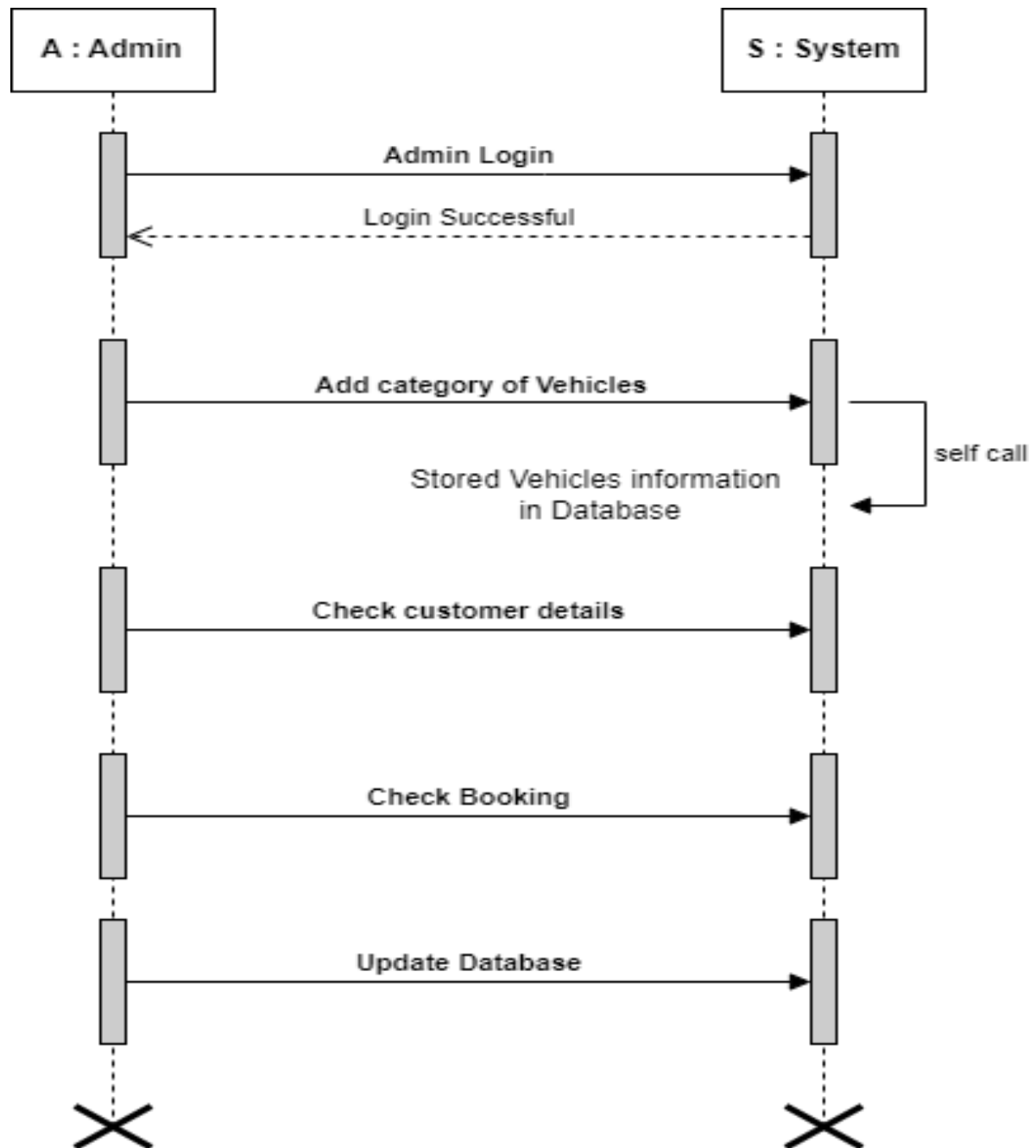


B) Object Diagram

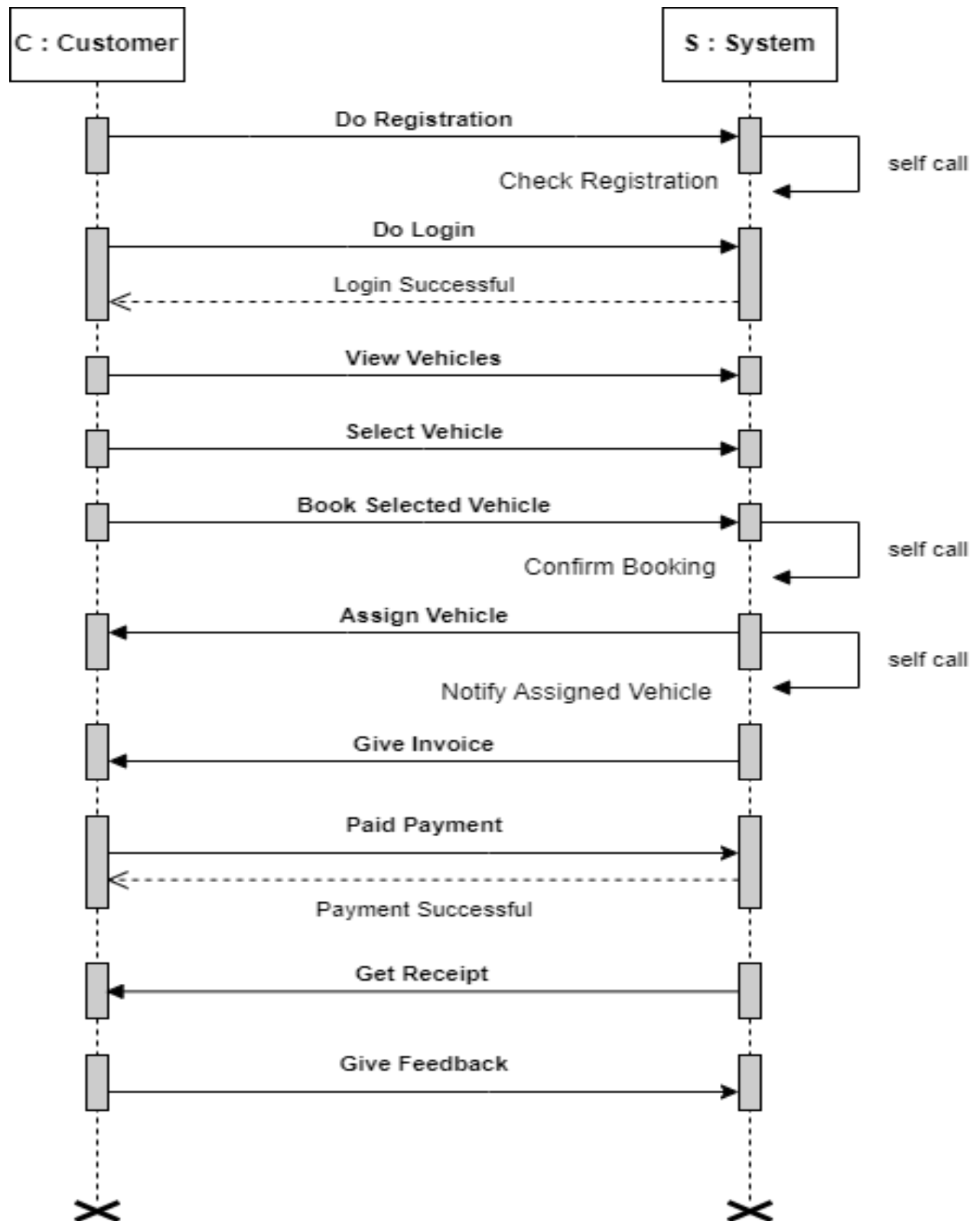


C) Sequence Diagram

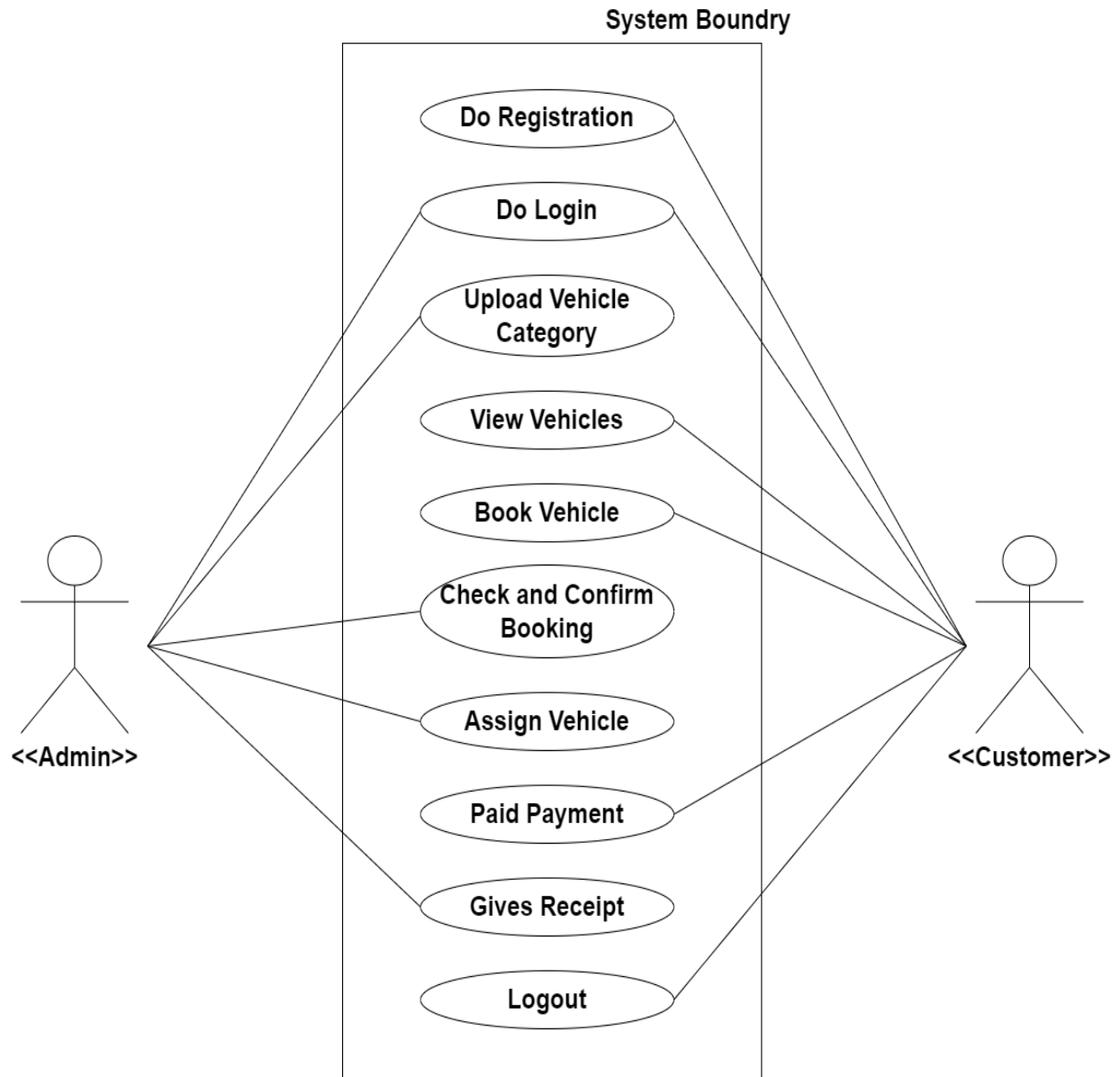
- Admin Sequence Diagram -



- **Customer Sequence Diagram –**

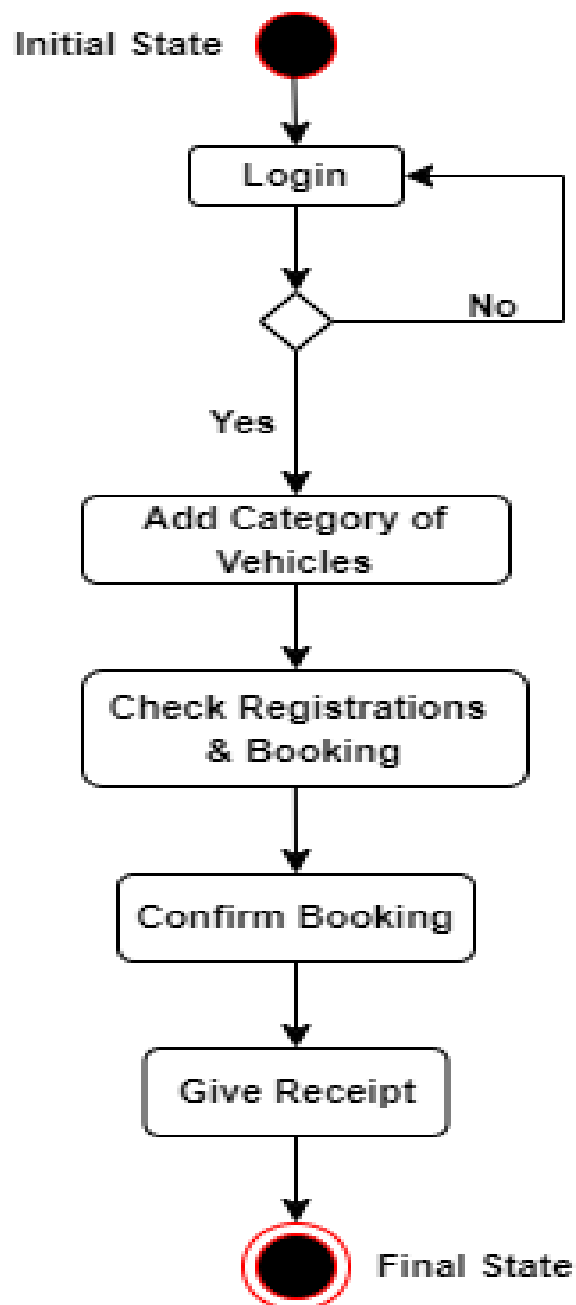


D) Use Case Diagram

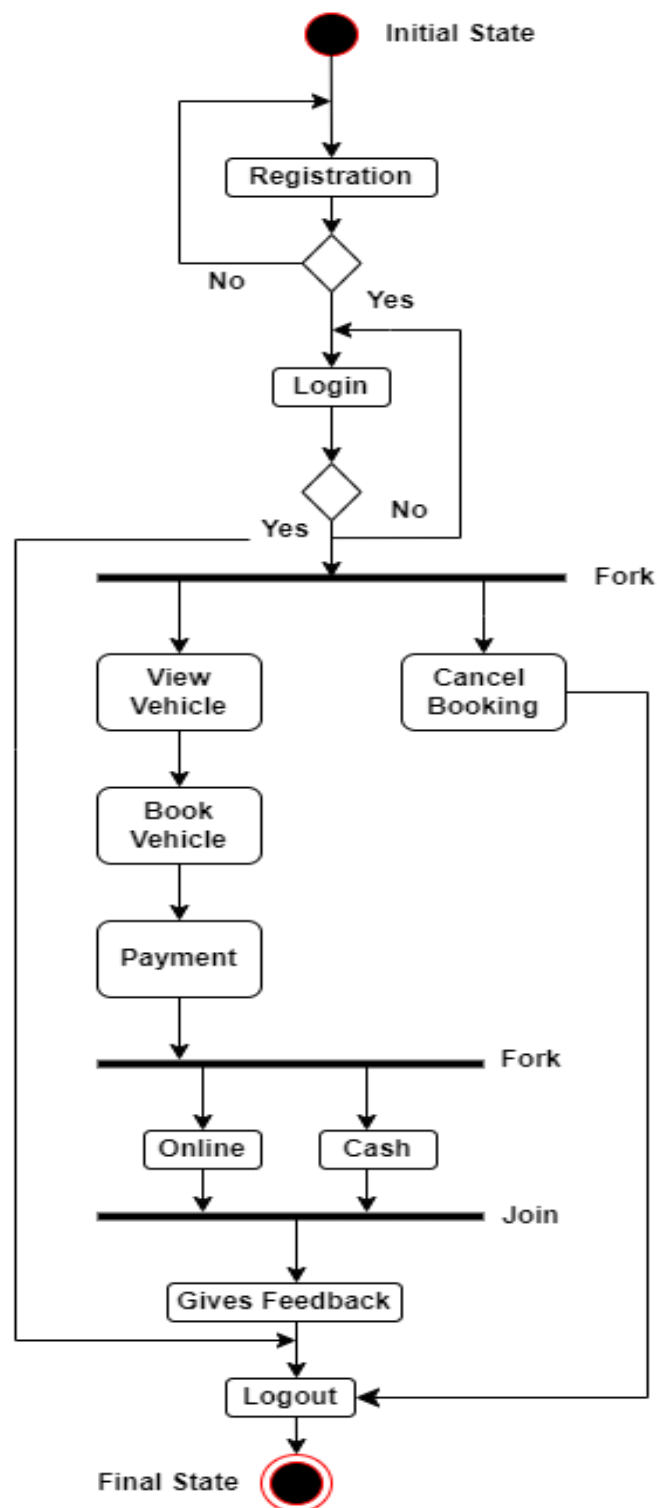


E) Activity Diagram

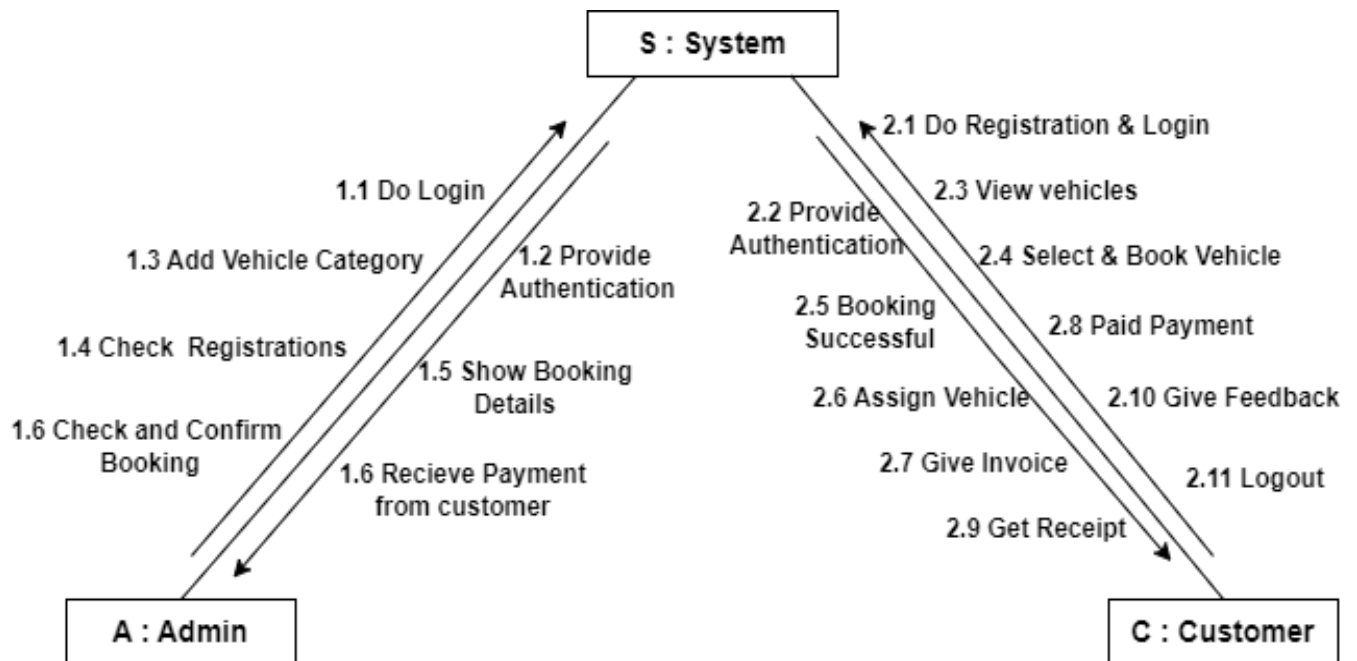
- Admin Activity Diagram –



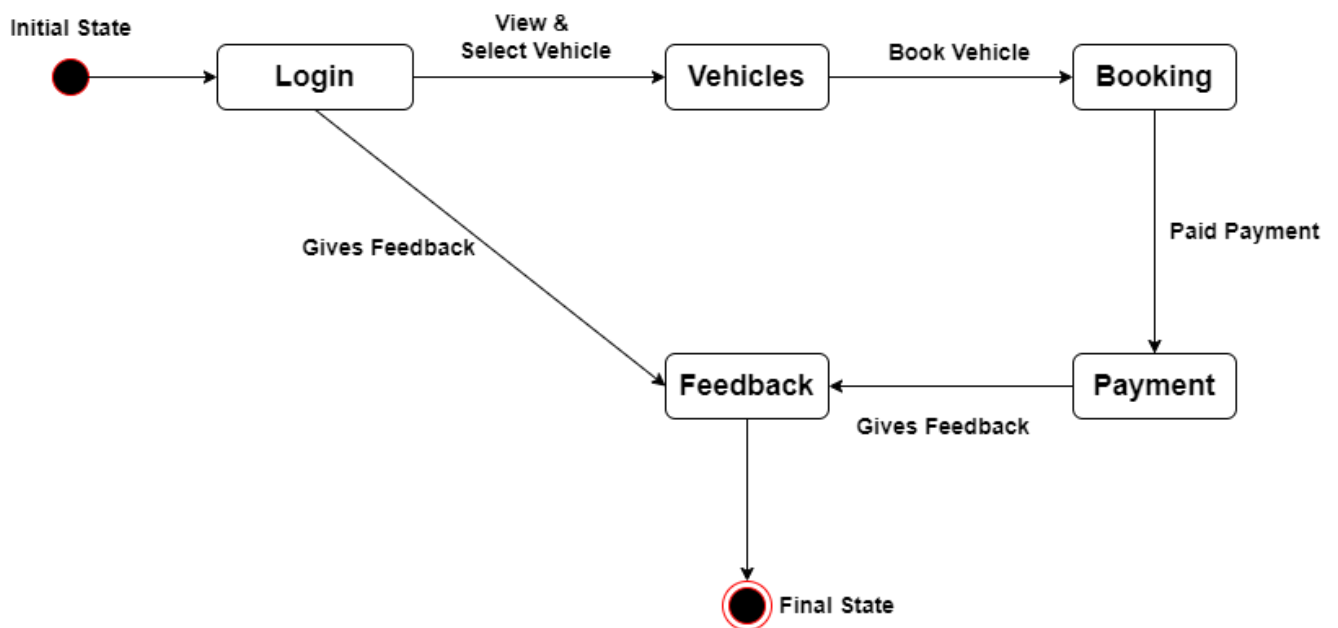
- **Customer Activity Diagram –**



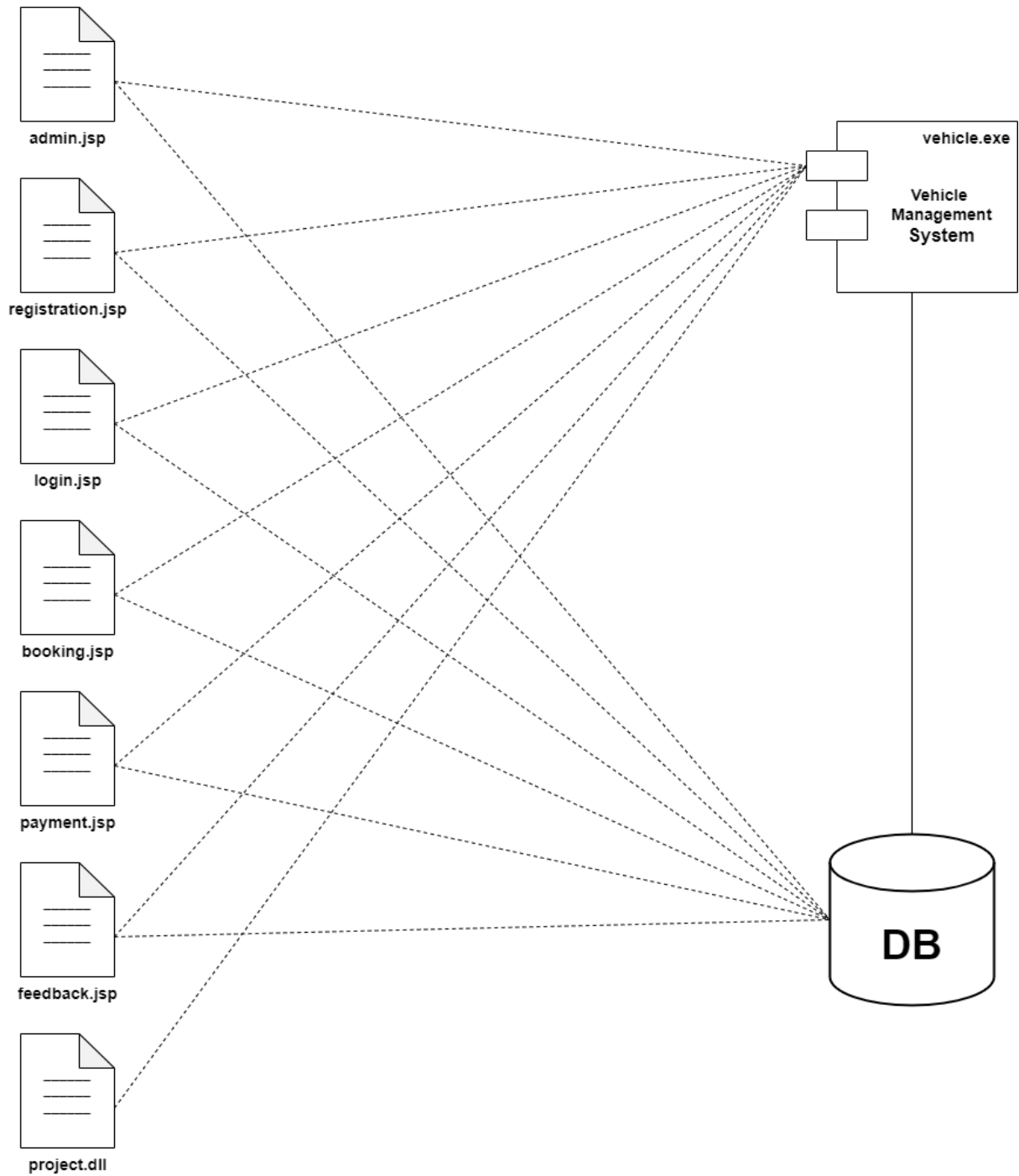
F) Collaboration Diagram



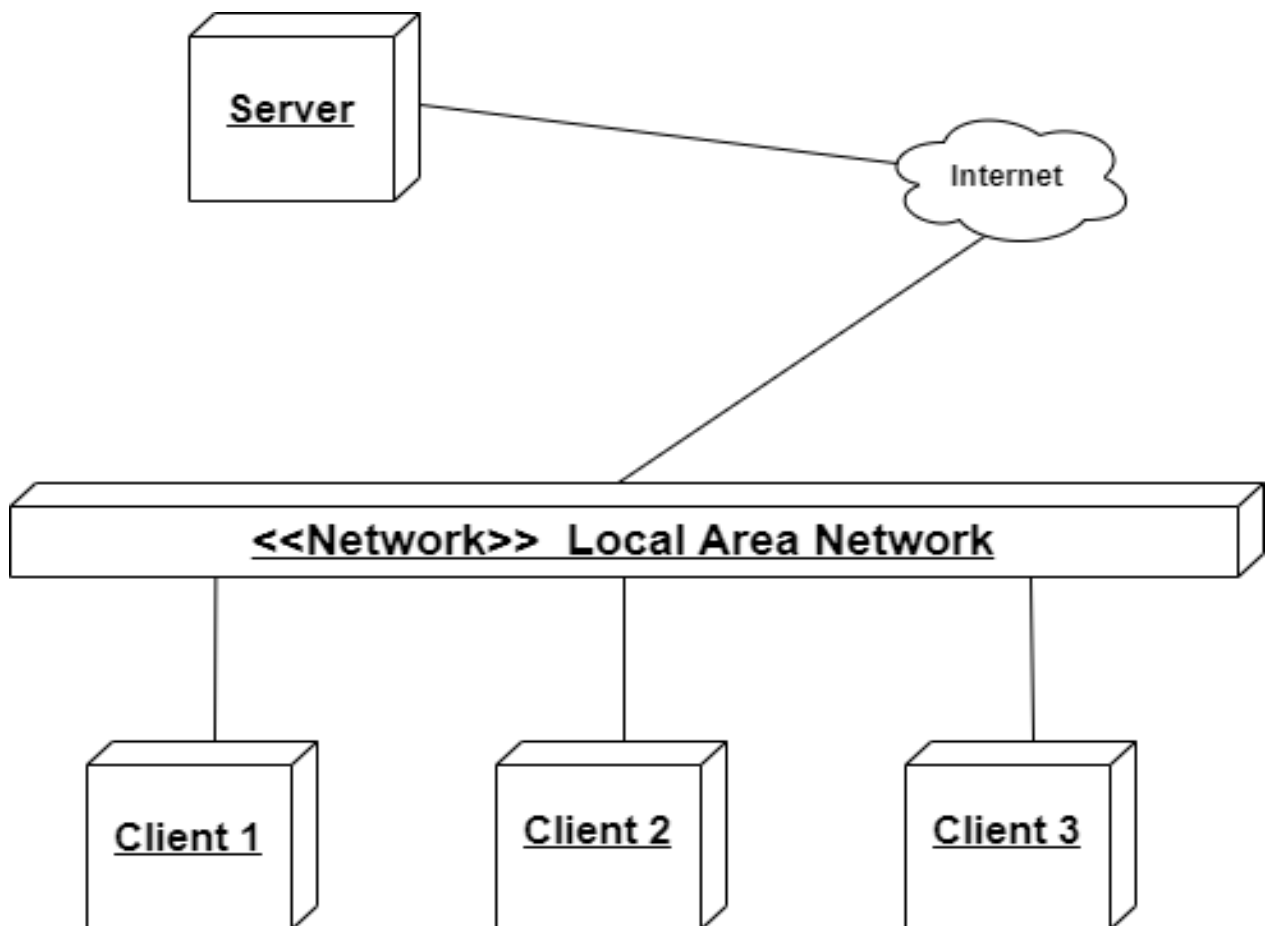
G) State Chart Diagram



H) Component Diagram



I) Deployment Diagram



➤ Data Dictionary

Registration Table

Field	Datatype	Null	Key	Default
rid	serial	No	Primary Key	-
rname	text	No	-	NULL
email	varchar(30)	No	Unique	NULL
password	varchar(30)	No	-	NULL

Customer Table

Field	Datatype	Null	Key	Default
cid	serial	No	Primary Key	-
cname	text	No	-	NULL
address	text	Yes	-	NULL
phone	varchar(15)	Yes	-	NULL

Vehicle Table

Field	Datatype	Null	Key	Default
vno	serial	No	Primary Key	-
vmodel	text	No	-	NULL
color	text	No	-	NULL
price	decimal(10,2)	No	-	NULL
cid	integer	No	Foreign Key	NULL

Booking Table

Field	Datatype	Null	Key	Default
bid	serial	No	Primary Key	-
cid	integer	No	Foreign Key	NULL
vno	integer	No	Foreign Key	NULL
date	date	No	-	NULL
status	varchar(20)	No	-	'Pending'

Payment Table

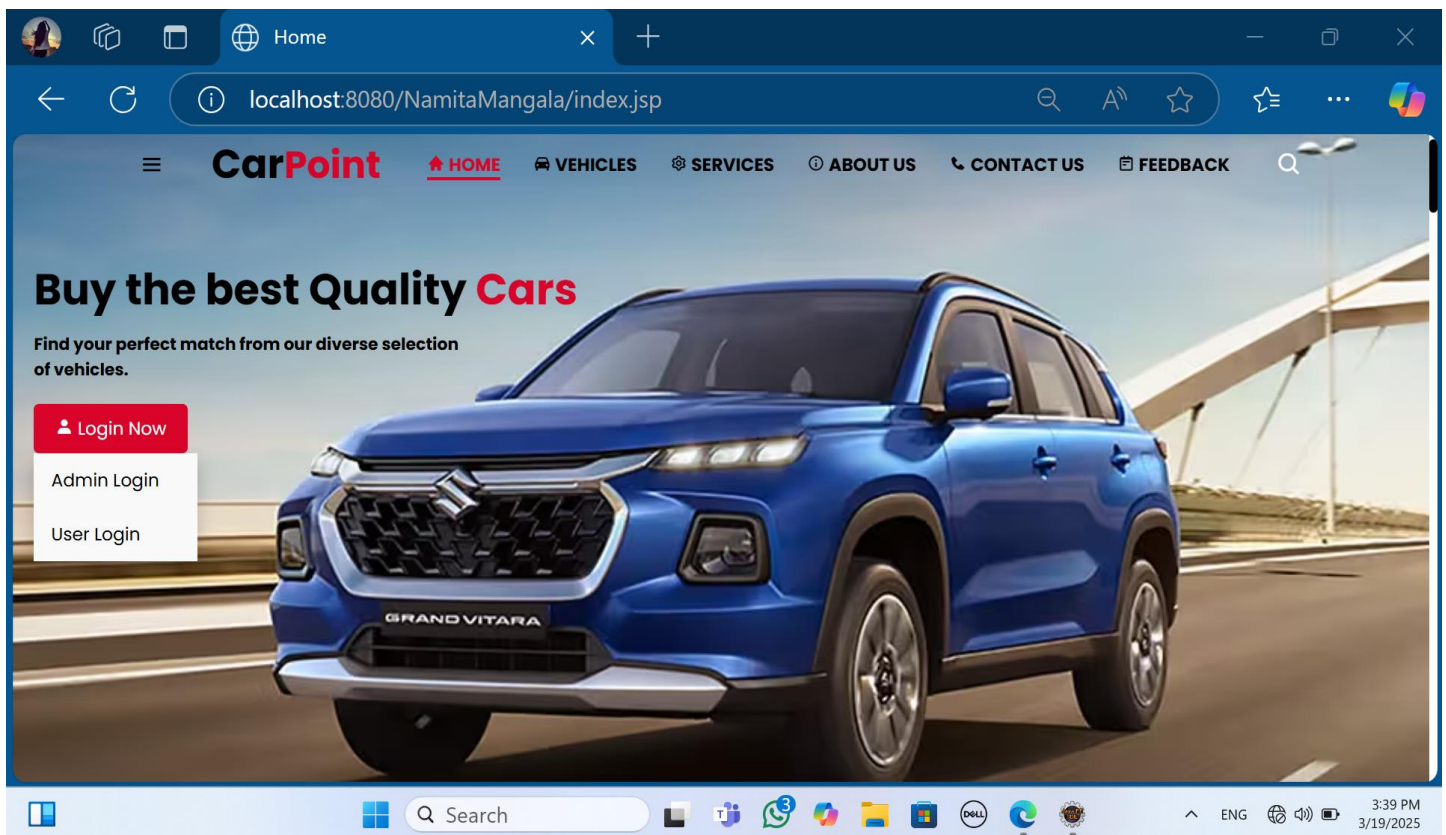
Field	Datatype	Null	Key	Default
pid	serial	No	Primary Key	-
ptype	text	Yes	-	NULL
pamount	decimal(10,2)	No	-	NULL
date	date	No	-	NULL
cid	integer	No	Foreign Key	-

Feedback Table

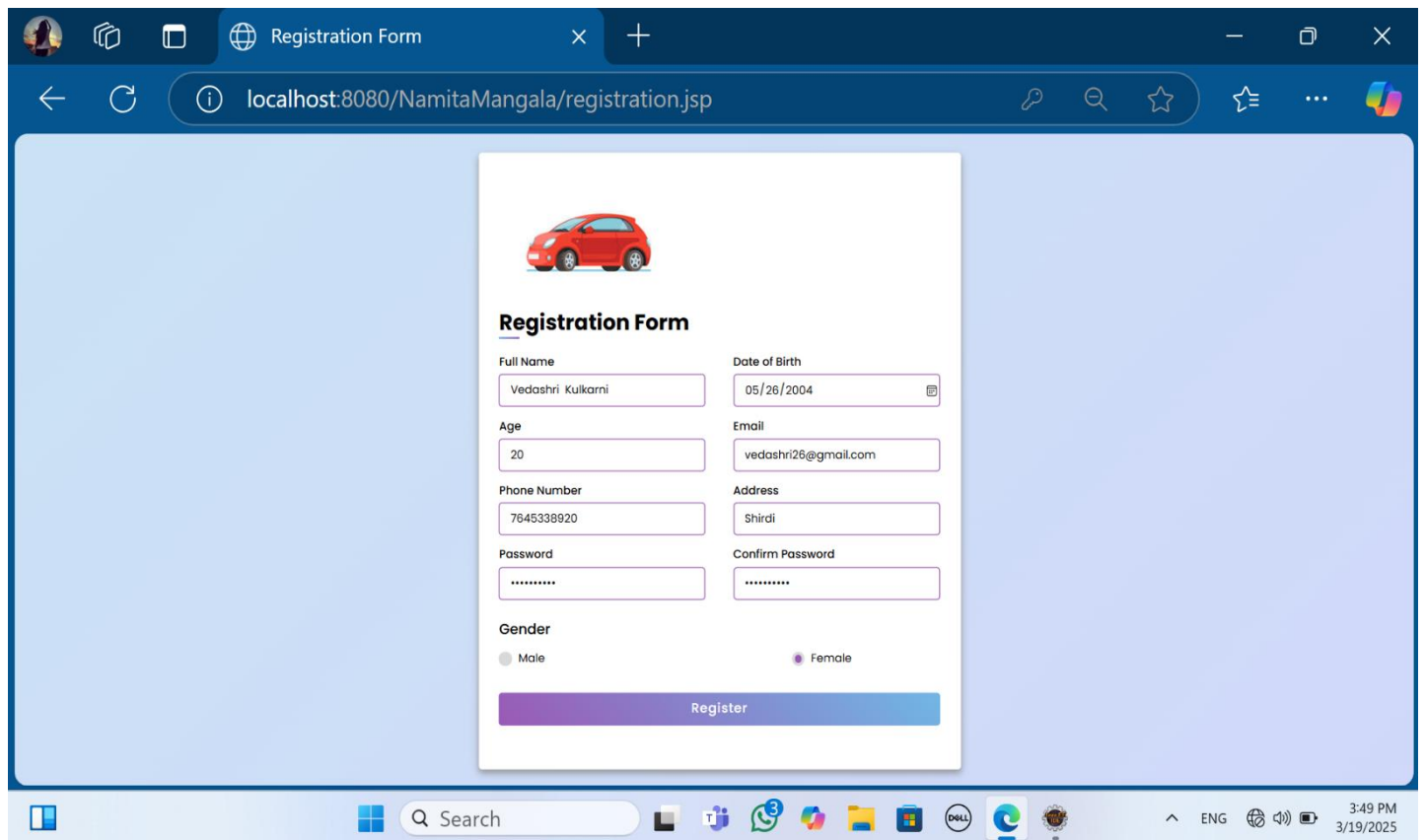
Field	Datatype	Null	Key	Default
fid	serial	No	Primary Key	-
name	text	No	-	NULL
date	date	Yes	-	NULL
rating	integer	No	-	NULL
response	text	Yes	-	NULL

➤ Input – Output Screens

- Home Page :



- **Registration Form Page :**



The screenshot displays a web browser window with a single tab titled "Registration Form". The address bar shows the URL "localhost:8080/NamitaMangala/registration.jsp". The form itself is centered on a light blue background and features a red car icon at the top. Below the icon, the title "Registration Form" is displayed. The form contains several input fields: "Full Name" (filled with "Vedashri Kulkarni"), "Date of Birth" (filled with "05/26/2004"), "Age" (filled with "20"), "Email" (filled with "vedashri26@gmail.com"), "Phone Number" (filled with "7645338920"), "Address" (filled with "Shirdi"), "Password" (filled with "*****"), and "Confirm Password" (filled with "*****"). There are radio buttons for "Gender" with "Male" selected and "Female" unselected. A blue "Register" button is positioned at the bottom of the form.

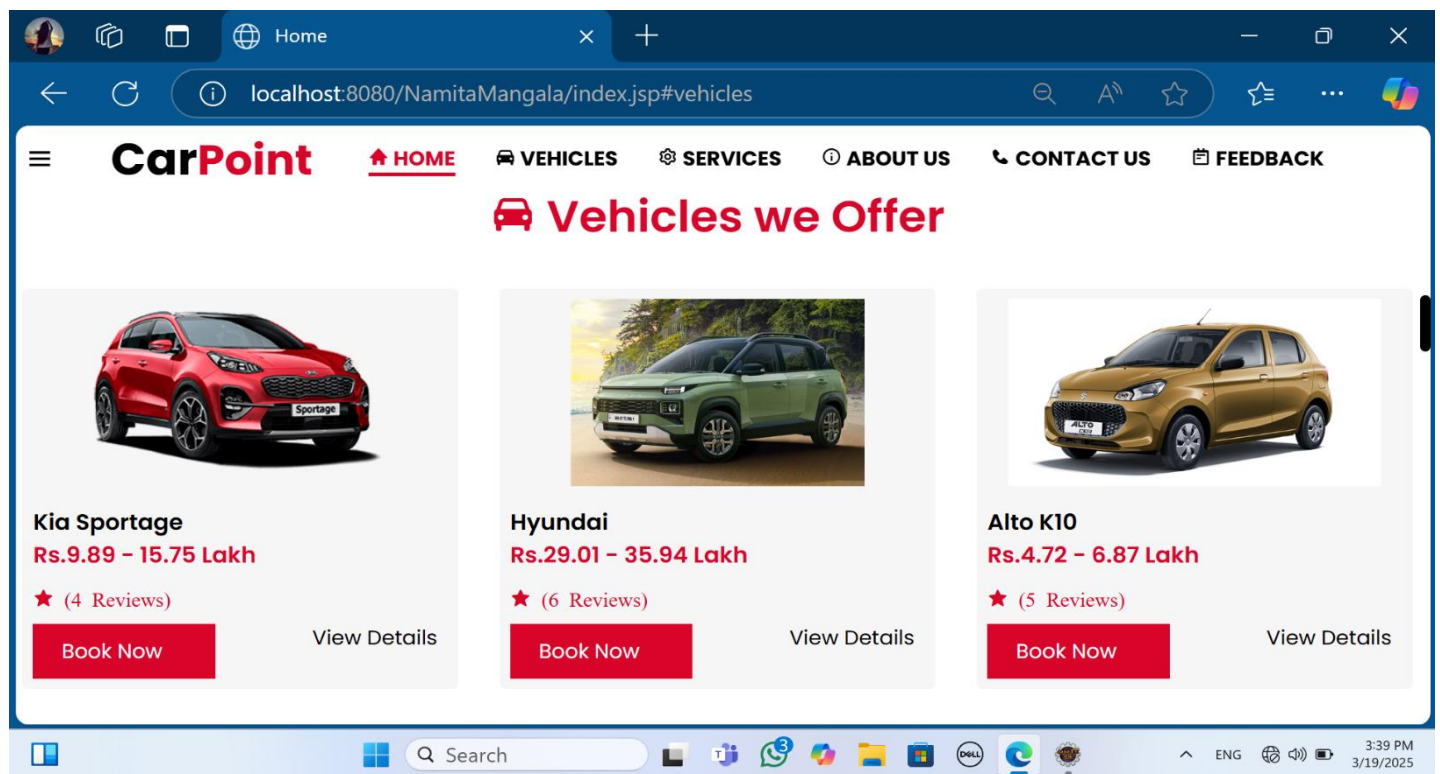
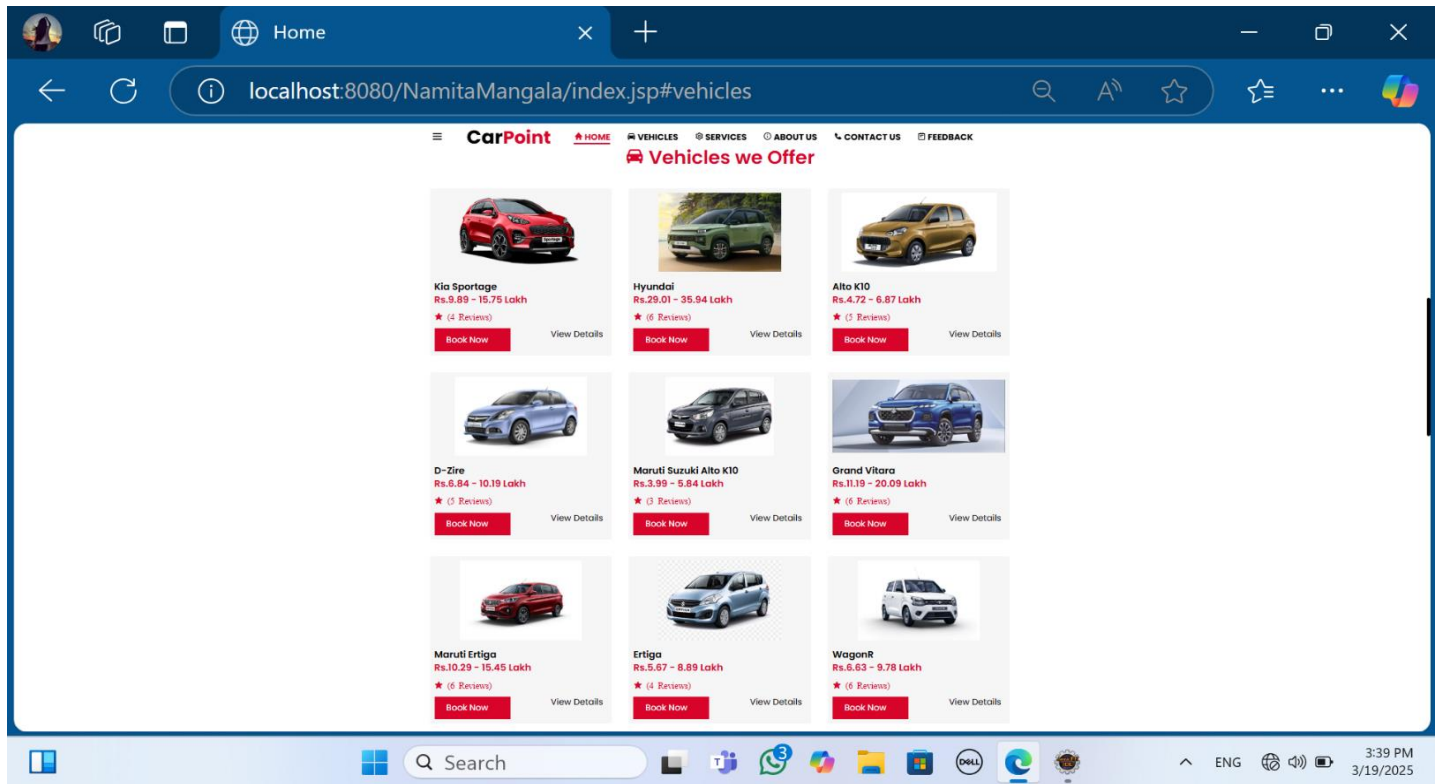
Field	Value
Full Name	Vedashri Kulkarni
Date of Birth	05/26/2004
Age	20
Email	vedashri26@gmail.com
Phone Number	7645338920
Address	Shirdi
Password	*****
Confirm Password	*****
Gender	Male

Register

- **User Login Form Page :**

The screenshot displays a web browser window with a single tab titled "Login Form". The address bar shows the URL "localhost:8080/NamitaMangala/user-login.jsp". The main content area features a "User Login Form" with a purple header. Below the header, there are two input fields: "Email Address" containing "vedashri26@gmail.com" and "Password" with masked characters. A "Forgot password?" link is positioned below the password field. A prominent purple "Login" button is centered below the inputs. At the bottom of the form, a link reads "Don't have an account? Register here". The Windows taskbar at the bottom shows the search bar, several application icons, and the system clock indicating 3:50 PM on 3/19/2025.

- Vehicles Page :



Home

CarPoint
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[VEHICLES](#)
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[CONTACT US](#)
[FEEDBACK](#)

Vehicles we Offer

Kia Sportage
Rs.9.89 - 15.75 Lakh
★ (4 Reviews)
[Book Now](#) [View Details](#)

Hyundai
Rs.29.01 - 35.94 Lakh
★ (6 Reviews)
[Book Now](#) [View Details](#)

Alto K10
Rs.4.72 - 6.87 Lakh
★ (5 Reviews)
[Book Now](#) [View Details](#)

D-Zire
Rs.6.84 - 10.19 Lakh
★ (5 Reviews)
[Book Now](#) [View Details](#)

Maruti Suzuki Alto K10
Rs.3.99 - 5.84 Lakh
★ (3 Reviews)
[Book Now](#) [View Details](#)

Grand Vitara
Rs.11.19 - 20.09 Lakh
★ (6 Reviews)
[Book Now](#) [View Details](#)

ENG
3:42 PM 3/19/2025

Home

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[ABOUT US](#)
[CONTACT US](#)
[FEEDBACK](#)

Vehicles we Offer

D-Zire
Rs.6.84 - 10.19 Lakh
★ (5 Reviews)
[Book Now](#) [View Details](#)

Maruti Suzuki Alto K10
Rs.3.99 - 5.84 Lakh
★ (3 Reviews)
[Book Now](#) [View Details](#)

Grand Vitara
Rs.11.19 - 20.09 Lakh
★ (6 Reviews)
[Book Now](#) [View Details](#)

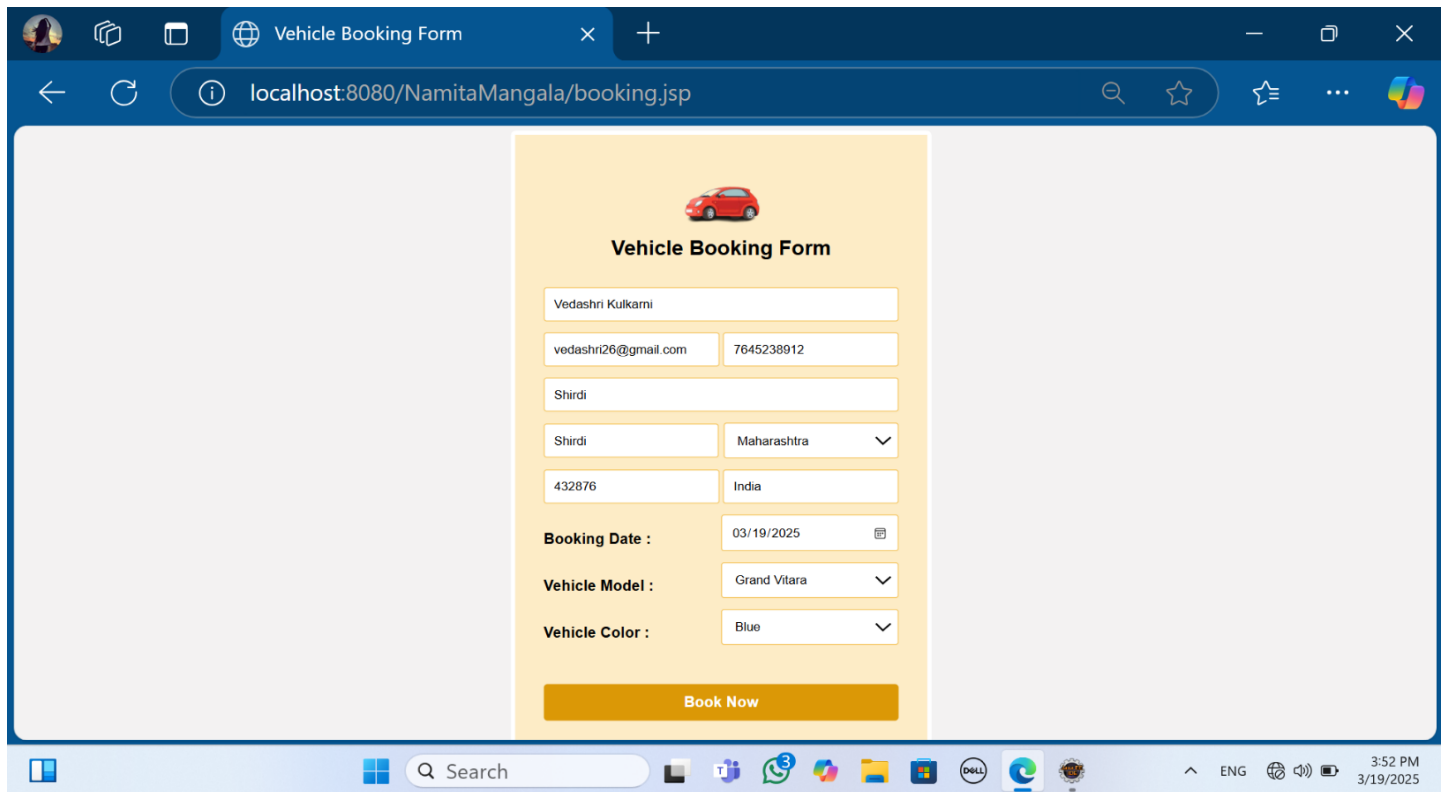
Maruti Ertiga
Rs.10.29 - 15.45 Lakh
★ (6 Reviews)
[Book Now](#) [View Details](#)

Ertiga
Rs.5.67 - 8.89 Lakh
★ (4 Reviews)
[Book Now](#) [View Details](#)

WagonR
Rs.6.63 - 9.78 Lakh
★ (6 Reviews)
[Book Now](#) [View Details](#)

ENG
3:42 PM 3/19/2025

- **Vehicle Booking Form Page :**



Vehicle Booking Form

Customer Name: Vedashri Kulkarni

Email: vedashri26@gmail.com | Phone: 7645238912

Location: Shirdi

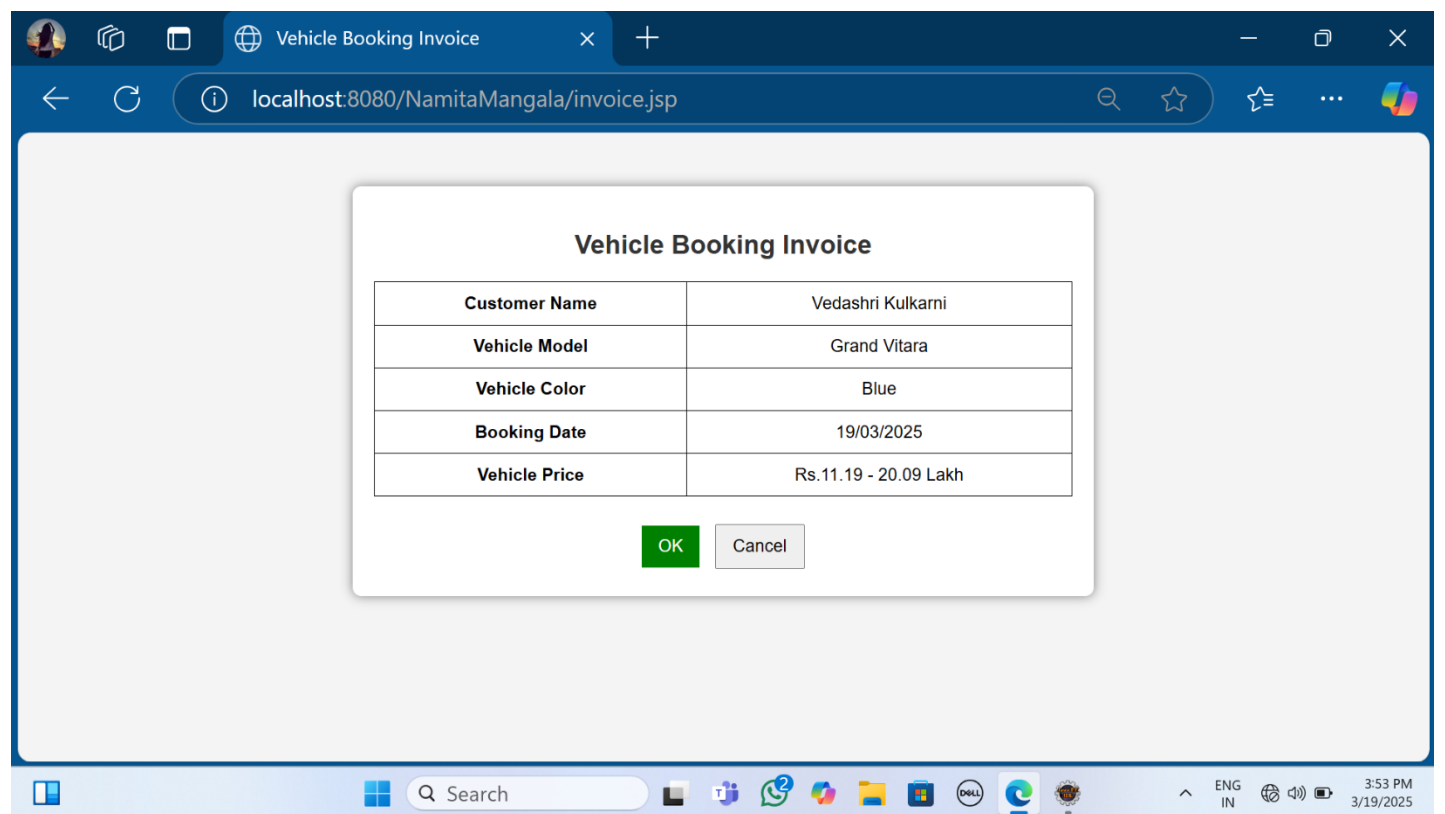
State: Maharashtra | Pincode: 432876 | Country: India

Booking Date: 03/19/2025

Vehicle Model: Grand Vitara

Vehicle Color: Blue

Book Now

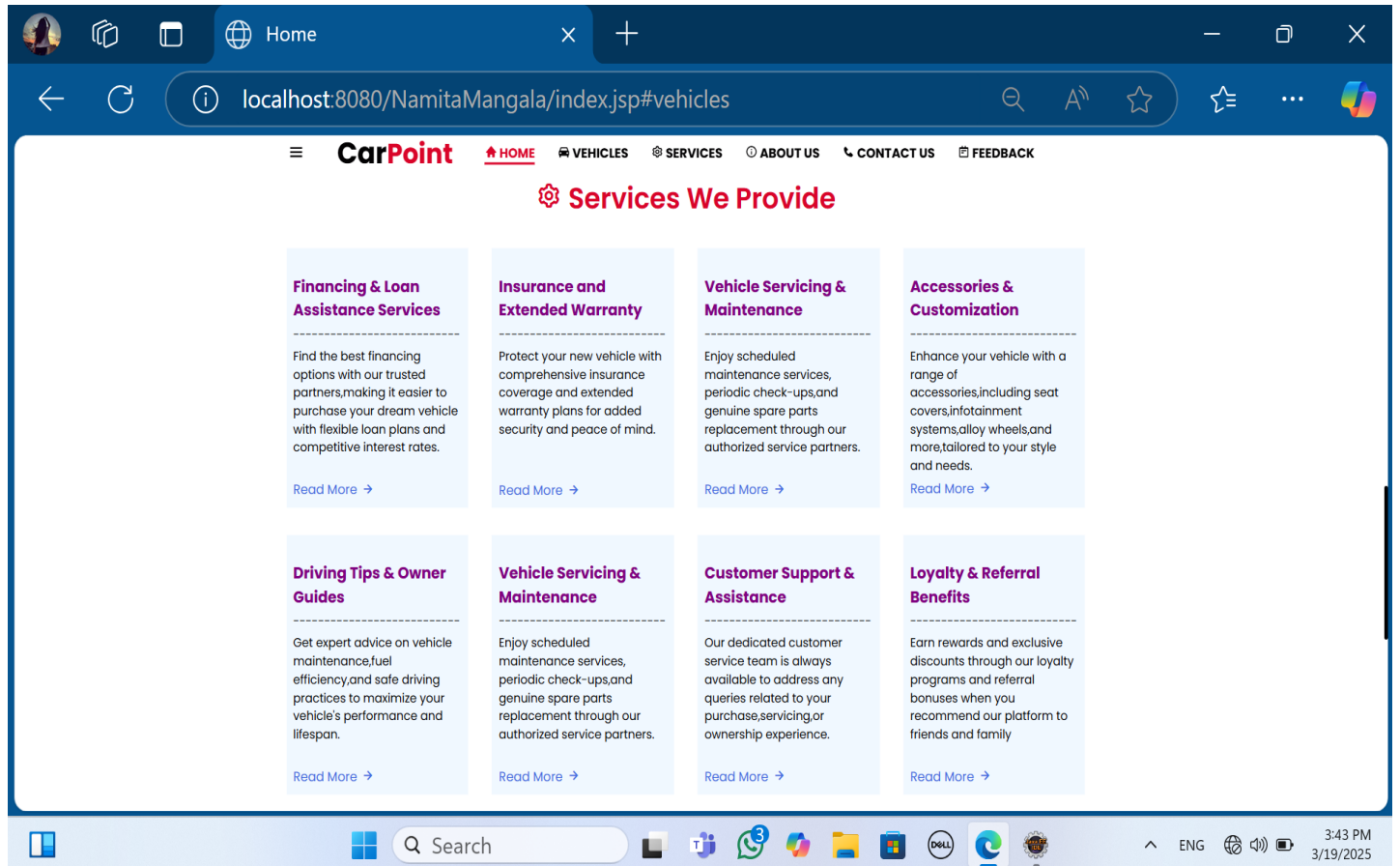


Vehicle Booking Invoice

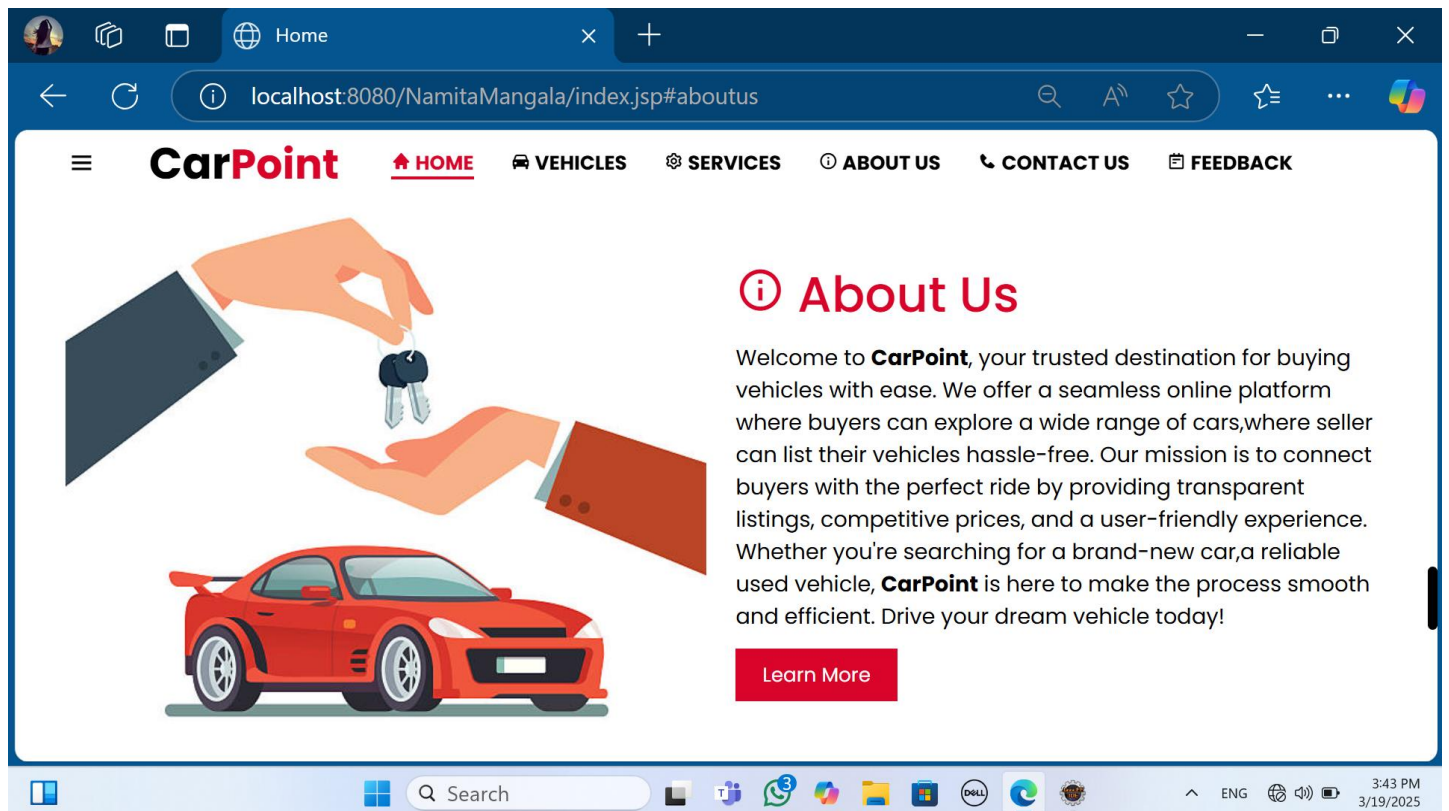
Customer Name	Vedashri Kulkarni
Vehicle Model	Grand Vitara
Vehicle Color	Blue
Booking Date	19/03/2025
Vehicle Price	Rs.11.19 - 20.09 Lakh

OK **Cancel**

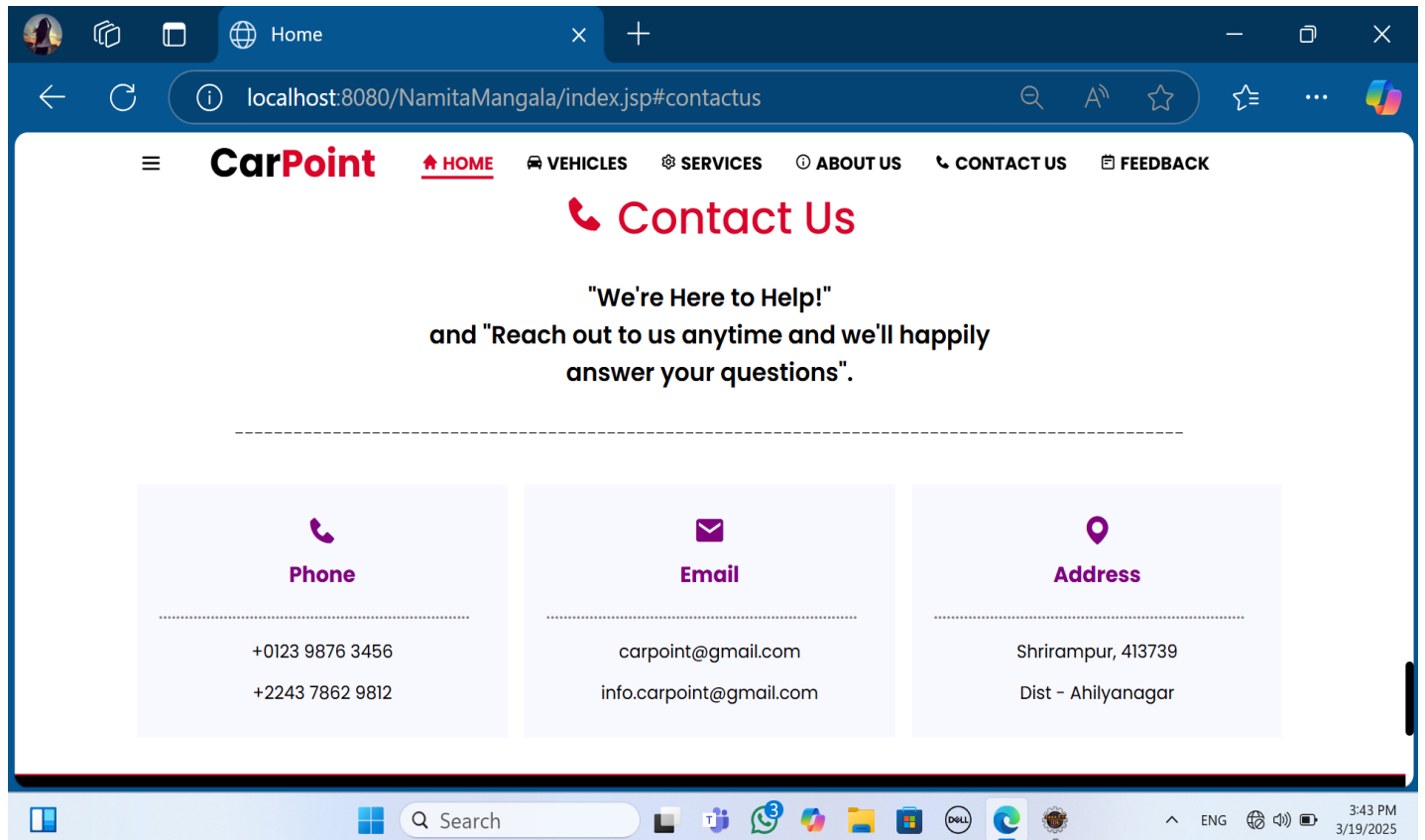
- **Services Page :**



- **About Us Page :**



- **Contact Us Page :**



- **Customer Feedback Form Page :**

Customer Feedback

Full Name :
Vedashri Kulkarni

Email Address :
vedashri26@gmail.com

Date :
03/25/2025

Rating :
★★★★★ - Excellent

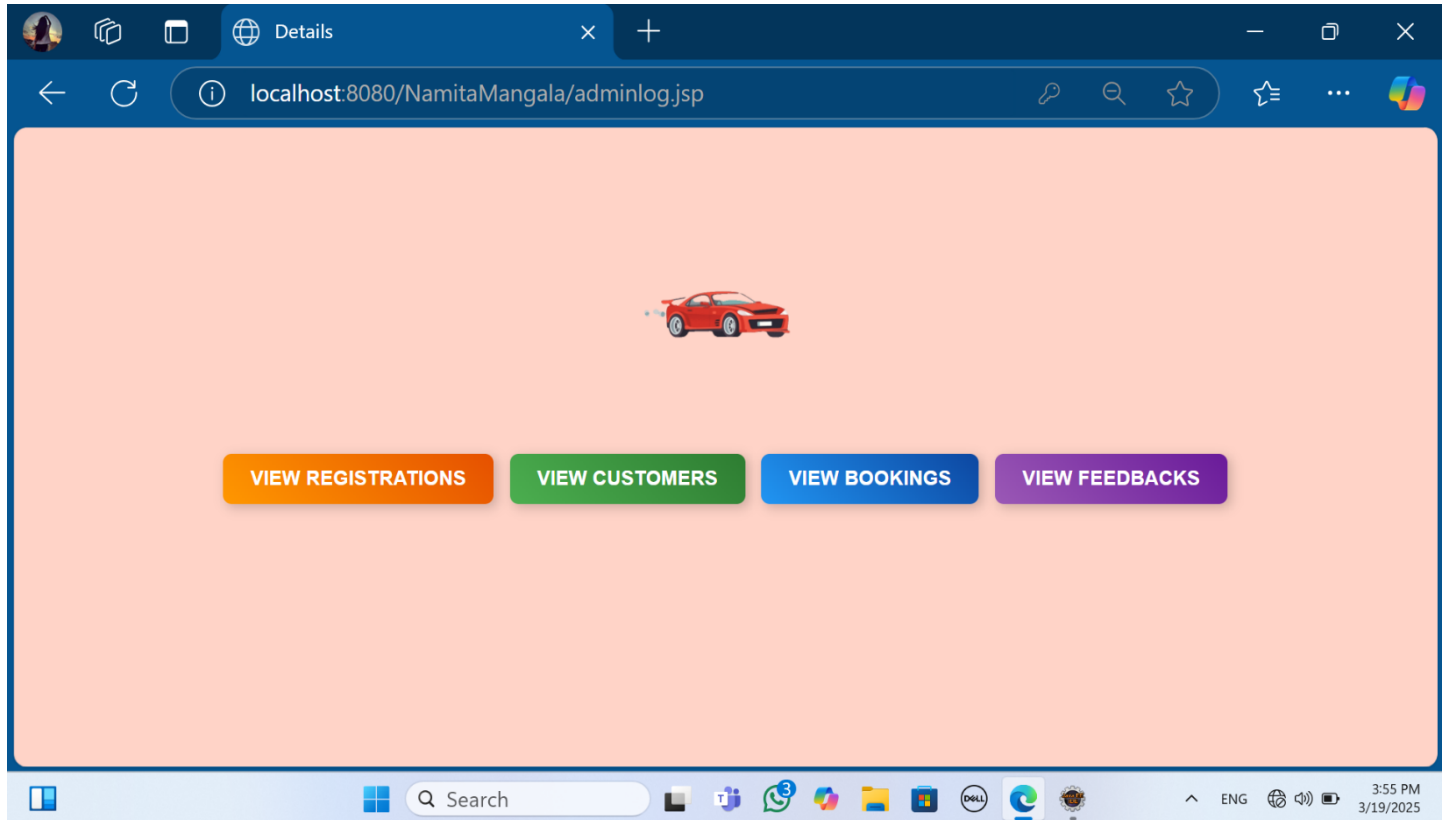
Response :
Excellent Cars 🍌

Submit Feedback

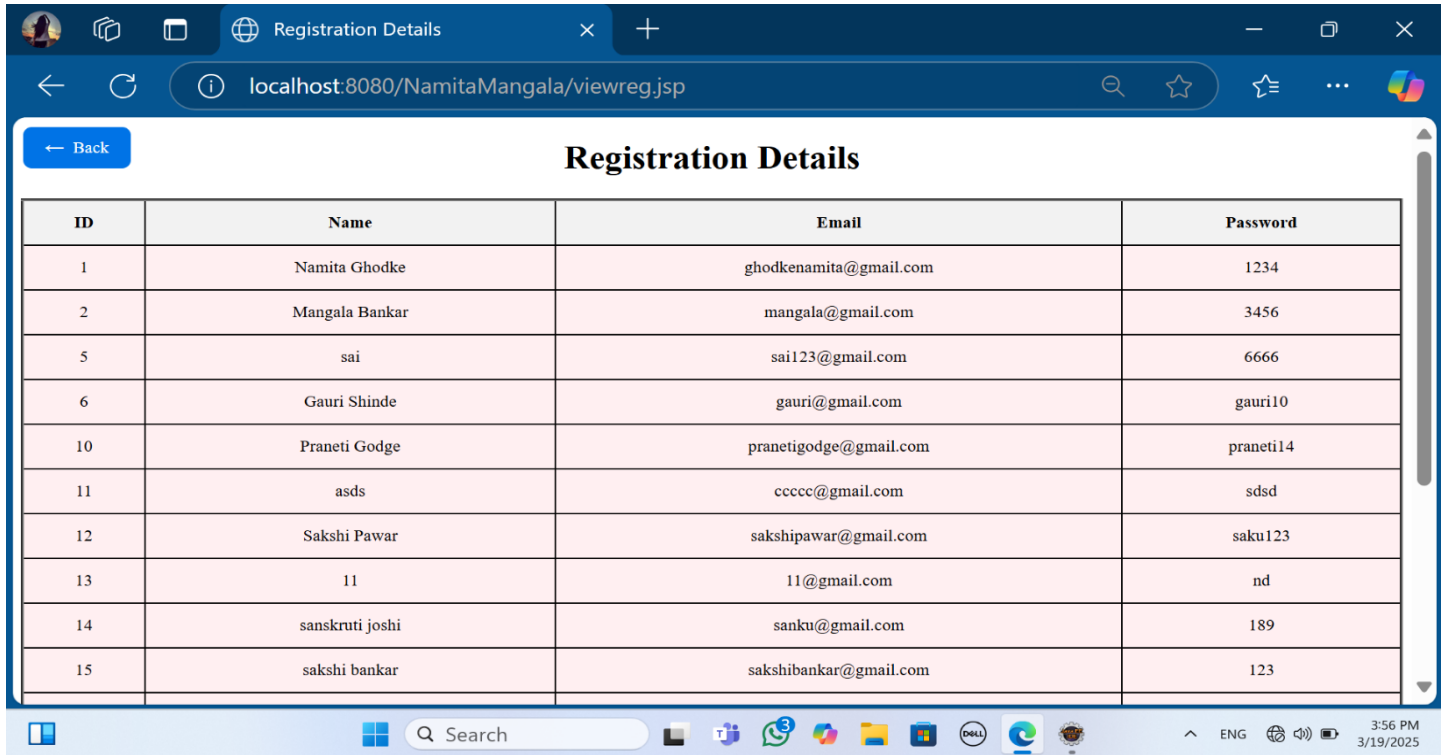
- **Admin Login Form Page :**

The screenshot displays a web browser window with a single tab titled 'Login Form'. The address bar shows the URL 'localhost:8080/NamitaMangala/admin-login.jsp'. The main content area features a light blue background with a central white login form. The form has a purple header with the text 'Admin Login Form'. Below the header, there are two input fields: 'Email Address' with the value 'Admin' and 'Password' with masked characters. A link labeled 'Forgot password?' is positioned below the password field. At the bottom of the form is a purple 'Login' button. The Windows taskbar at the bottom shows the search bar, several application icons, and the system clock indicating 3:55 PM on 3/19/2025.

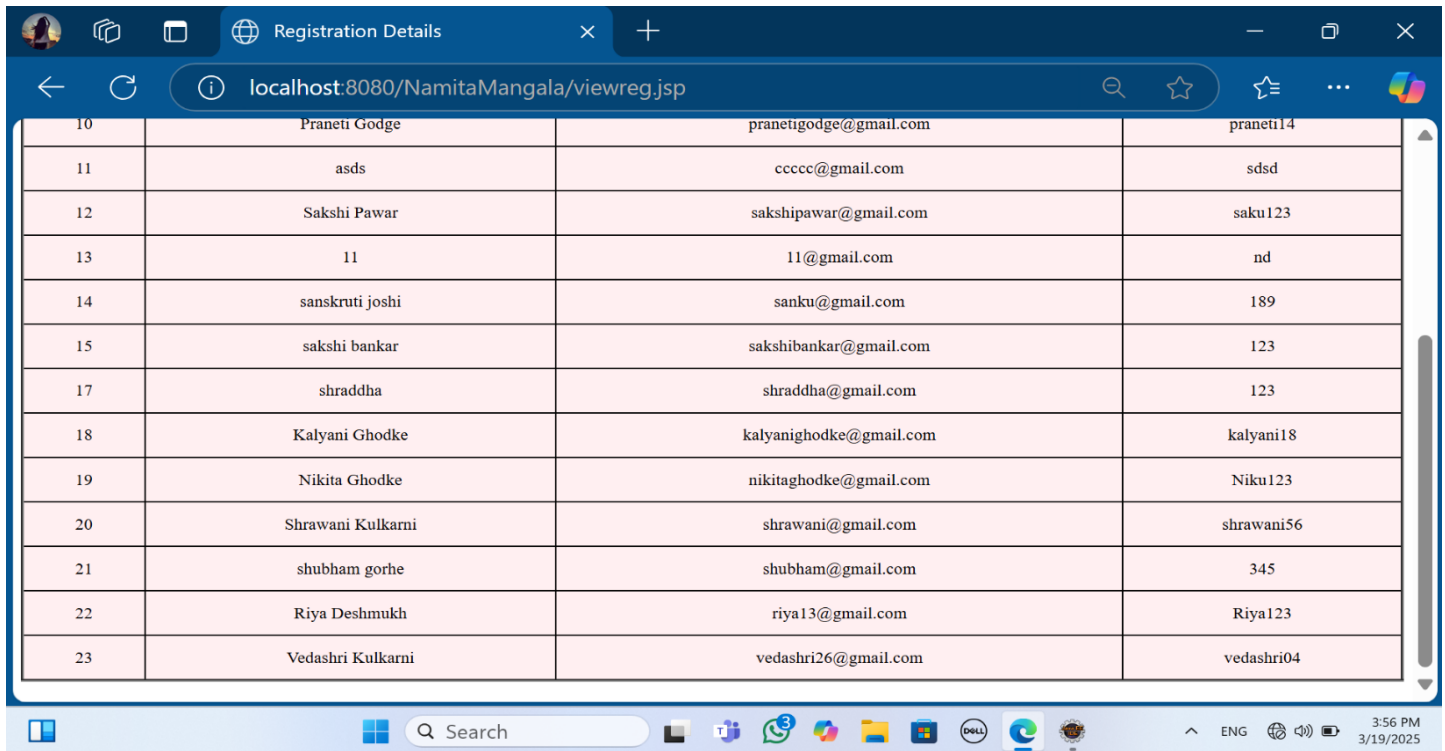
- **Admin Check Details Page :**



• Registration Details Page :



ID	Name	Email	Password
1	Namita Ghodke	ghodkenamita@gmail.com	1234
2	Mangala Bankar	mangala@gmail.com	3456
5	sai	sai123@gmail.com	6666
6	Gauri Shinde	gauri@gmail.com	gauri10
10	Praneti Godge	pranetigodge@gmail.com	praneti14
11	asds	ccccc@gmail.com	sdsd
12	Sakshi Pawar	sakshipawar@gmail.com	saku123
13	11	11@gmail.com	nd
14	sanskriti joshi	sanku@gmail.com	189
15	sakshi bankar	sakshibankar@gmail.com	123



10	Praneti Godge	pranetigodge@gmail.com	praneti14
11	asds	ccccc@gmail.com	sdsd
12	Sakshi Pawar	sakshipawar@gmail.com	saku123
13	11	11@gmail.com	nd
14	sanskriti joshi	sanku@gmail.com	189
15	sakshi bankar	sakshibankar@gmail.com	123
17	shraddha	shraddha@gmail.com	123
18	Kalyani Ghodke	kalyanighodke@gmail.com	kalyani18
19	Nikita Ghodke	nikitaghodke@gmail.com	Niku123
20	Shrawani Kulkarni	shrawani@gmail.com	shrawani56
21	shubham gorhe	shubham@gmail.com	345
22	Riya Deshmukh	riya13@gmail.com	Riya123
23	Vedashri Kulkarni	vedashri26@gmail.com	vedashri04

- **Customer Details Page :**

← Back

Customer Details

ID	Name	Address	Phone	Action
8	sakshi bankar	khairi	4254236563	Remove
9	Shrawani Kulkarni	Wadala Mahadeo	8855828056	Remove
10	shubham gorhe	nashik	63767367	Remove
11	Riya Deshmukh	Bandra	9834561289	Remove
12	Vedashri Kulkarni	Shirdi	7645238912	Remove

• Booking Details Page :

← Back

Booking Details

Booking ID	Customer Name	Vehicle Model	Vehicle Color	Address	Date	Status	Action
11	Sanskriti Joshi	Scorpio	blue	shrirampur	2025-03-06	Confirmed	<button>Confirm</button> <button>Cancel</button>
13	Shrawani Kulkarni	Hyundai	black	Wadala Mahadeo	2025-03-18	Confirmed	<button>Confirm</button> <button>Cancel</button>
2	namita	Hyundai	white	shrirampur	2025-03-19	Confirmed	<button>Confirm</button> <button>Cancel</button>
6	om	Grand Vitara	blue	mumbai	2025-02-15	Pending	<button>Confirm</button> <button>Cancel</button>
12	sakshi bankar	Scorpio	black	khairi	2025-03-05	Pending	<button>Confirm</button> <button>Cancel</button>
10	shreya	Swift	red	pune	2025-03-07	Pending	<button>Confirm</button> <button>Cancel</button>
9	Vidya	Scorpio	white	andheri	2025-03-13	Pending	<button>Confirm</button> <button>Cancel</button>
14	shubham gorhe	Grand Vitara	black	nashik	2025-03-13	Pending	<button>Confirm</button> <button>Cancel</button>

← Back

Booking Details

6	om	Grand Vitara	blue	mumbai	2025-02-15	Pending	<button>Confirm</button> <button>Cancel</button>
12	sakshi bankar	Scorpio	black	khairi	2025-03-05	Pending	<button>Confirm</button> <button>Cancel</button>
10	shreya	Swift	red	pune	2025-03-07	Pending	<button>Confirm</button> <button>Cancel</button>
9	Vidya	Scorpio	white	andheri	2025-03-13	Pending	<button>Confirm</button> <button>Cancel</button>
14	shubham gorhe	Grand Vitara	black	nashik	2025-03-13	Pending	<button>Confirm</button> <button>Cancel</button>
16	Vedashri Kulkarni	Grand Vitara	blue	Shirdi	2025-03-19	Pending	<button>Confirm</button> <button>Cancel</button>
8	Sham	Grand Vitara	black	Nashik	2025-03-19	Pending	<button>Confirm</button> <button>Cancel</button>
15	Riya Deshmukh	Ertiga	blue	Bandra	2025-03-19	Pending	<button>Confirm</button> <button>Cancel</button>
4	Mangala Bankar	Hyundai	white	Padhegaon	2025-03-26	Pending	<button>Confirm</button> <button>Cancel</button>
7	Ram	Maruti Suzuki	grey	nashik	2025-04-19	Pending	<button>Confirm</button> <button>Cancel</button>

- **Feedback Details Page :**

← Back

Feedback Details

ID	Customer Name	Date	Rating	Response
1	Namita	2025-03-19	4	Nice
3	Shruti Holkar	2025-03-26	4	Nice
4	Amruta Ghodke	2025-03-18	4	Very Nice
5	Shrawani Kulkarni	2025-03-18	5	Luxurious Cars
6	Riya Deshmukh	2025-03-26	5	Wonderful Cars!
7	Vedashri Kulkarni	2025-03-25	5	Excellent Cars 🍌

Windows taskbar: Search, 3:57 PM, 3/19/2025

➤ **Advantages and Disadvantages**

Advantages :

- Real Time Tracking
- Improved Efficiency
- Cost Reduction
- Automated Alerts and Reminders
- Reduces the time consumption
- Decreases the paper and labour work
- Manage the entire process

Disadvantages :

- High Initial Cost
- Data Security Risks
- Ongoing Maintenance Costs
- Privacy Concerns

➤ **Future Enhancement**

- This is not the overall description about the vehicle selling system.
- We can connect the sms alert system to the proposed system which will enhance the efficiency of the system.
- The email system can be use to connect the user and admin.
- Improve Safety with automatic notifications.
- Allow test drive for customer.
- Provide nearby locations for service stations and also charging points for electric vehicles to new customer in the area.
- Provides instance customer support.

➤ **Conclusion**

This project is design for the purpose to provide a seamless platform for buying and selling vehicles, enhancing user experience through automation, secure transactions, and real-time vehicle information.

Customer can easily book vehicle without wastage of time as per their expectation.

In this system Admin or Seller can easily upload the vehicle details and advertise their vehicle by saving the time. Admin can view booking vehicle details, customer details and update the database.

➤ **Bibliography**

❖ **Reference Books –**

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 - T.Y.B.Sc (Computer Science) Book
- Herbert Schildt Java : The Complete Reference, McGraw Hill, 11th Edition, 2019
- Programming with Java : A Primer - E. Balagurusamy
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- Web Programming : E. Balagurusamy

❖ **Websites and Links -**

- www.google.com
- www.javatpoint.com
- www.geeksforgeek.com
- www.stackoverflow.com